Opacity as Commitment: Fragile Financial Markets under Delegated Trading*

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Abstract

We study a model of delegated asset management in which a fund manager cares not only about trading profits but also about her reputation, captured by investors' perceptions of the manager's skill. The fund is intentionally made opaque by the manager, who communicates noisy information about her trading strategy to clients. This opacity generates two self-fulfilling equilibria: a benchmark equilibrium and a "flow-driven" equilibrium. In the benchmark equilibrium, the manager prioritizes fund performance, and her trading intensity is set at the profit-maximizing level. In contrast, in the flow-driven equilibrium, the manager trades more aggressively in order to appear skilled, thereby building reputation among investors and attracting larger fund flows, while the fund exhibits poor performance. Our results imply that (1) fund opacity and reputation concerns can trigger market fragility and increase inequality in managers' compensation; (2) fund managers with different skill levels adopt different trading styles; and (3) skill in active management does not necessarily translate into superior performance. Our model also shows that highly skilled fund managers are more likely to choose opaque fund structures, as opacity makes it difficult to communicate trading strategies to investors and weakens incentives for reputation-driven risk-taking. In this sense, fund opacity serves as a commitment device that allows skilled managers to pursue profit-driven trading strategies rather than engaging in flashy risk-taking aimed at signaling high skill.

JEL classification: G11, G12, G14, G23

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1 Introduction

Obfuscation is common in investor-directed fund communications. Even when pursuing identical strategies, managers describe them in markedly different and often unnecessarily complex ways. For example, two funds that both track the S&P 500 describe essentially identical objectives with strikingly different clarity: Schwab states in a single sentence, "The fund's goal is to track the total return of the S&P 500 Index," while Deutsche describes it in 60 words, embedding the same objective in technical language.[1] Similar patterns arise in fund names: labels such as "Dynamic Opportunity Fund," "Flexible Growth Strategy," "Opportunistic Equity Fund," or "Smart Beta" often describe portfolios of large-cap stocks that closely resemble the S&P 500. This pattern is puzzling, as standard signaling theory suggests that managers seeking to convey skill and attract capital should prefer greater transparency rather than opacity. The literature emphasizes competitive concerns or proprietary information as managers' obfuscation incentive, yet these arguments are less compelling when strategies are easily inferred by sophisticated market participants. This raises a central question: why would managers make information harder for their own investors to interpret, and what are the incentives and market consequences when fund opacity becomes a manager's strategic choice?

To address these questions, we study a model of delegated asset management in which a fund manager, privately informed about her skill, raises capital from investors and trades risky assets with market makers in the financial market à la Kyle (1985). We define skill as the manager-specific ability to uncover more information about asset payoffs than the market: the more skilled the manager, the larger the mispricing she can identify and the greater the trading profits she can potentially earn. To attract fund flows, the manager communicates information about her risky position (portfolio) to

[1]Deutsche's S&P 500 index fund is described as "The fund seeks to provide investment results that, before expenses, correspond to the total return of common stocks publicly traded in the United States, as represented by the Standard & Poor's 500 Composite Stock Price Index (S&P 500 Index). The fund invests for capital appreciation, not income; any dividend and interest income is incidental to the pursuit of its objective."

investors. Investors, in turn, use this fund information to infer the manager's skill and decide on capital allocation to the fund. Importantly, the manager strategically chooses the precision of this communication and may intentionally obfuscate fund information. The more opaque the fund information, the more difficult for investors to back out the manager's underlying skill.

The model yields two stable, self-fulfilling equilibria regarding the fund's trading strategy. One of the equilibria is similar to that of Kyle (1985), where the manager employs the profit-maximizing trading strategy. In the other, which we call the "flow-driven" equilibrium, the manager trades the risky asset more aggressively to appear skilled and to attract large fund flows. This strategy increases trading volume, price volatility, and market liquidity but undermines the fund's trading profits by revealing too much private information to the market maker. Intuitively, the manager can boost investors' perception of her skill by placing a large order, either buy or sell, because such an action makes her appear like a skilled manager who has identified an asset with significant mispricing. In equilibrium, investors accurately anticipate the manager's strategy, so their learning is not manipulated; nevertheless, the manager may take excessive risk following the market's belief. To illustrate the mechanism, suppose that investors *believe* that the manager trades more aggressively than the profit-maximizing level. If she trades less aggressively to increase trading profits, investors (incorrectly) infer that the manager is less skilled. To avoid the reduction in fund flows, she conforms to investors' belief and indeed trades very aggressively.

The self-fulfilling property of our equilibria implies that a mere shift in investor beliefs, unaccompanied by changes in fundamentals, can trigger a transition from the standard Kyle-like equilibrium to the flow-driven one, where price volatility is high and trading performance is relatively poor. We interpret this as *fragility* in financial markets.[2] Key to our multiple equilibria and fragility results are the manager's conflicting motives for revealing skill: on one hand, she wishes to hide her skill from the market maker to earn trading

[2]As in Greenwood and Thesmar (2011), we define a market as fragile if it is susceptible to non-fundamental demand shifts, such as those caused by changes in investor beliefs.

profits, but on the other hand, she wants to show it off to investors to establish high reputation and attract large fund flows.

The relative strength of these hiding and showing-off motives is governed by how sensitive fund flows are to the information the manager communicates about her portfolio. For this reason, the degree of fund opacity becomes a key parameter determining whether fragility arises. When fund information is highly obfuscated, investors find it difficult to infer skill from fund information, and capital allocations respond only weakly to the manager's trading strategy. In this case, the showing-off motive is weak, and the manager optimally follows the hiding motive, trading close to the Kyle (1985) benchmark and maximizing performance. By contrast, when fund information is highly transparent, small changes in trading intensity substantially affect investors' inferences and, in turn, fund flows. The benefit from inflating risky positions to attract large flows becomes substantial, leading to the flow-driven equilibrium with excessive risk-taking. At intermediate levels of opacity, these competing forces are balanced, giving rise to self-fulfilling multiple equilibria and financial fragility.

Our results shed light on how funds opacity relates to managers' trading styles and contribute to financial fragility. The model predicts that highly opaque funds adopt low-intensity, profit-driven strategies, while those with transparent information engage in aggressive, risk-seeking behavior to gain reputation and fund flows. Funds with intermediate degree of opacity, facing multiple equilibria, may switch between profit-seeking and flow-driven strategies based on market beliefs, thereby contributing to market fragility. This behavior aligns with real-world observations during financial crises, such as momentum crashes and sudden reversals, which occur even without fundamental shocks.

The model has implications for compensation in the finance industry. In the reputation-driven equilibrium, traders' wages exhibit greater inequality among high- and low-skilled traders compared to those in the Kyle-like equilibrium, as they trade aggressively and reveal their skill more than they do in the Kyle-like equilibrium. This allows the recruiter to better assess each trader's skill and, therefore, to offer wages that more accurately reflect their abilities. This result is consistent with the findings by Philippon and Reshef (2012) and Célérier and Vallée (2019), who document increasing returns to talent and their significant contribution to income inequality in the finance industry.

The aggressive trade in the reputation-driven equilibrium also causes traders to reveal excessive information to the market maker, resulting in poorer trading performance than in the Kyle-like equilibrium. Consequently, high-skilled traders in the reputation-driven equilibrium may underperform low-skilled traders following the Kyle-like equilibrium. This result offers a novel explanation for the puzzling relationship (or the lack thereof) between trader skill and persistent performance, as documented and analyzed in the literature (e.g., Busse, Goyal, and Wahal, 2010; Berk and Green, 2004).

In Section **??**, we study traders' endogenous skill acquisition. We show that, due to the multiplicity of equilibria, each trader may intentionally acquire a "middling" level of skill, which is too high or too low compared to the level she would acquire if she knew which type of equilibrium would prevail. Ex-post, in the Kyle-like equilibrium, where profits are prioritized over reputation, the acquired skill level is insufficient. Conversely, in the reputation-driven equilibrium, where reputation outweighs profit motives, the skill level is excessive. Ex-ante, without knowing which equilibrium will emerge, the trader optimally chooses a middling skill level to prepare for both scenarios. This finding aligns with real-world observations that some finance professionals tend to pursue broader business qualifications rather than specializing in areas like advanced computer science or pursuing higher academic degrees such as a PhD.

Our paper is related to the literature on fragility in financial markets. Cespa and Vives (2015) is most closely related, as they also find multiple equilibria in an asymmetric information trading model. The key ingredients of their model are short-term investor horizons and persistent liquidity trading, which generate strategic complementarities. In Froot, Scharfstein, and Stein (1992), strategic complementarities among short-term traders are also important for the emergence of multiple equilibria. Unlike these studies, our model does not feature short horizons; instead, we obtain fragility from traders' forward-looking concerns about future recruitment opportunities. In Cespa and Foucault (2014), learning from other assets amplifies shocks and causes liquidity crashes. Although the mechanism is different, our model can also result in liquidity crashes when the economy switches between equilibria due to non-fundamental shocks.

Our paper also connects to the literature on trading frenzies, such as Veldkamp (2006) and Goldstein, Ozdenoren, and Yuan (2013), in which strategic complementarities play a key role. In Veldkamp (2006) frenzies originate from complementarities in information markets, while in Goldstein, Ozdenoren, and Yuan (2013) strategic complementaries, triggered by a feedback effect from financial markets to a firm's real investment, lead to coordinated aggressive trading. By contrast, in our reputation-driven equilibrium, each trader trades aggressively out of fear of being perceived as less skilled by the recruiter who believes that she would trade aggressively.

Lastly, our work contributes to the broader understanding of the role of asymmetric information about skills in financial markets. DiMaggio (2015), Malliaris and Yan (2021), and Bijlsma, Boone, and Zwart (2018) derive implications for the risk-taking behavior of agents when ability is private information but is reflected in performance. We also link trader skill to private information and asset markets; however, unlike these papers, our focus is on deriving pricing implications. Guerrieri and Kondor (2012) show that career concerns amplify price volatility, similar to our findings. In Dasgupta (2006), career concerns lead to churning (trading without private information), and in Dasgupta and Prat (2008), they lead to herding (ignoring private information and trading like everyone else). These three papers are closely related to ours in that they study career concerns and endogenous asset prices, but they do not analyze fragility, which is the main focus of our paper.

The rest of the paper proceeds as follows. Section 2 presents a baseline model with exogenous reputation concerns, and Section 3 studies the equilibrium. Section **??** endogenizes reputation concerns by introducing a two-period model and presents the implications of our results. Section **??** analyzes traders' skill acquisition. The Online Appendix contains all proofs for the theoretical results.

2 Model

This section presents a one-period trading model à la Kyle (1985) with delegated asset management.

2.1 Setting

The economy consists of a *fund manager*, a competitive *market maker*, a *noise trader*, and $n \geq 2$ of *investors*. All players are risk neutral. The manager leverages her skill (defined later) to identify a single risky asset with potential mispricing, raises capital from investors, and allocates it between the risky asset and a risk-free asset with a zero interest rate.[3]

2.1.1 Skill

The manager can uncover more information about the asset's payoff than the market maker, and we interpret it as her *skill*. At the beginning of the period, the manager draws private information s from a normal distribution with mean zero and variance ω_s. Leveraging this information, she identifies an asset with payoff $\delta = \bar{\delta} + s$, where $\bar{\delta}$ is a publicly known mean payoff. Since the market maker does not observe s, her prior expectation of the asset's payoff is $\mathrm{E}[\delta] = \bar{\delta} + \mathrm{E}[s] = \bar{\delta}$, which is biased by s from the informed manager's perspective. As the market maker would set a price $p = \mathrm{E}[\delta] = \bar{\delta}$ in the absence of trades, s can be viewed as the asset's "potential mispricing."[4] Indeed, as

[3] We abstract away from information acquisition by individual investors and assume that they invest through a manager due to the lack of informational advantages. This assumption would be supported by, for example, information- or skill-acquisition costs, where only the fund manager acquires the skill at lower costs due to economies of scale (e.g., Gârleanu and Pedersen, 2018).

[4] s is not the *actual* mispricing because the market maker will take into consideration the presence of the informed manager; thus, p will deviate from $\bar{\delta}$ in equilibrium.

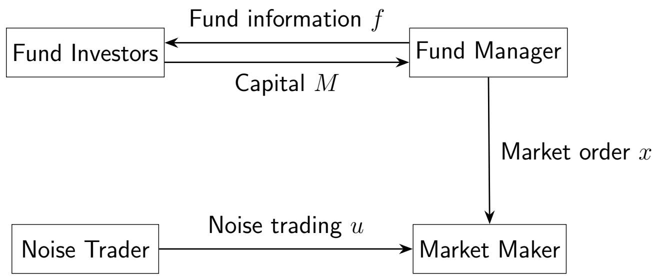


Figure 1: Capital Allocation and Investment

shown later, the manager profits by buying the asset if $s > 0$ (underpriced) and selling it if $s < 0$ (overpriced); the larger the $|s|$, the larger the profit. In light of this, we measure the manager's *realized* asset-selection skill by $|s|$, while the variance ω_s serves as an *ex-ante* measure of the average skill.[5,6]

2.1.2 Capital Allocation and Trading

We model the manager's trading decision and investors' capital allocation as a simultaneous game with rational expectations, as illustrated in Figure 1.

Manager's tradign decision. Upon discovering the mispricing in the asset (s), the fund manager chooses the portfolio allocation between the risky and the safe asset. This action is represented by a market order for $x \in \mathbb{R}$ units of the risky asset, submitted to the market maker in the financial market. The manager decides on x by incorporating the reactions of the financial market (i.e., the asset price) and the capital inflow from her investors, as described in the following.

Investors' capital allocation. Each investor chooses the amount of capital to invest into the fund. Investor i's capital allocation is denoted by $m_i \geq 0$,

[5]This formalization of skill aligns with findings in the empirical studies that attribute fund managers' performance to stock-picking skills (e.g., Wermers, 2000; Kosowski, Timmermann, Wermers, and White, 2006).

[6]The definition of the average skill follows from $\mathrm{E}[|s|] = \omega_s\sqrt{\frac{2}{\pi_0}}$, where π_0 denotes the circular constant.

and the total fund flow or the asset under management (AUM) of the fund is defined by $M \equiv \sum_{i=1}^{n} m_i$. The fund's fee structure is proportional to the total fund proceeds: the manager takes a $\phi \in (0,1)$ fraction of the proceeds, while the remaining $1-\phi$ fraction is distributed to the investors in proportion to their contributions to the fund, i.e., m_i/M.[7]

Financial market. In the financial market, the market maker sets the asset price, p, based on the incoming order flow, $x+u$, where u represents a random market order from the noise trader and follows a normal distribution with mean zero and variance ω_u. Due to competition, the market maker sets a semi-strong efficient price conditional on $x+u$:[8]

$$p = \mathrm{E}[\delta | x+u]. \tag{2.1}$$

2.1.3 Fund Opacity

At the fund-raising stage, the manager disseminates a private noisy signal about her trading strategy to investors, defined as $f = x + e$, where e represents a normally distributed noise term with mean zero and variance ω_e.[9] This signal is referred to as the *fund information*, and the noise variance of fund information, ω_e, is used as the measure of *fund opacity*. It describes the limited precision of the information that investors receive about the manager's true trading strategy. A higher ω_e means that disclosed fund information is noisier, making the strategy harder to verify or reverse engineer. This reflects the idea that managers may deliberately obfuscate details by being vague, selective, or technically complex in their descriptions, while still engaging with investors

[7] Introducing a fixed fee does not change our qualitative results, as long as the performance-based fee also exists, $0 < \phi < 1$.

[8] Assume that, on the equilibrium path, one competitive market maker trades the asset. As in Kyle (1985), competitively many market makers exist off the equilibrium path, ensuring a break-even price in the equilibrium.

[9] We assume that f is privately communicated between the manager and investors and is not observable to the market maker. Allowing the market maker to observe a noisy signal about x, however, does not affect the main qualitative results, as long as this signal differs from f, since the core results rely on the sensitivity of p to s being an increasing function of the manager's trading intensity.

to raise capital and to comply with disclosure rules.[10] Conversely, a lower ω_e corresponds to more transparent communication, allowing investors to form sharper beliefs about the manager's actions. Since the manager chooses her trading strategy x based on the private signal s, the fund information f is informative about the manager's skill and the fund's return. Thus, fund investors use f to decide on their capital allocations. In what follows, we first consider the baseline model with exogenous opacity, while Section 4 endogenizes ω_e by allowing the manager to strategically choose fund opacity.

2.1.4 Maximization Problems and Equilibrium

The fund's profit from the risky investment is represented by $\pi \equiv (\delta - p)x$, so that the total fund proceeds, including those from the safe asset, are

$$y = \pi + M. \tag{2.2}$$

Therefore, given the realized skill s, the fund manager maximizes her expected fee income by controlling her investment strategy x:

$$\max_{x} \mathrm{E}\left[\phi y | s\right]. \tag{2.3}$$

Conversely, conditional on the fund information f, investor i maximizes her expected return from delegating capital management:[11]

$$\max_{m_i \geq 0} \mathrm{E}\left[\frac{m_i}{M}(1-\phi)y - m_i | f\right]. \tag{2.4}$$

[10]For example, funds that largely track simple benchmark strategies are sometimes marketed using complex labels such as "dynamic opportunity," "flexible growth," or "adaptive allocation," which convey sophistication while revealing little about actual portfolio construction. Similarly, managers may emphasize broad investment philosophies without clarifying how these translate into concrete portfolios.

[11]The fund manager may have an incentive to make a false report about y to her investors. Namely, by reporting $\tilde{y}$, she pays out $(1-\phi)\tilde{y}$ to investors and captures $y-(1-\phi)\tilde{y}$ on her own. To prevent this intentional misreporting, we assume that each investor can verify x after the trading stage. Alternatively, we may assume costly verification: investors collectively pay the verification cost C to back out x from the fund information in the end of the trading period. As long as C positively depends on the degree of fund opacity ω_e, our qualitative results remain the same.

Note that the first term represents the expected payment from the fund, and the second term is the cost of capital.

Definition 1. *The equilibrium is defined by the market maker's pricing strategy p, the fund manager's trading strategy x, and investors' capital allocations $\{m_i\}_{i=1}^n$, such that, (i) p satisfies* (2.1) *given $x+u$, (ii) x solves (2.3) given s, p, and $\{m_i\}_{i=1}^n$, and (iii) for all i, m_i solves (2.4) given p, f, and $\{m_j\}_{j\neq i}$.*

3 Equilibrium

In this section, we search for a linear equilibrium in which the manager's market order x is linear in her skill s, and the market maker's pricing strategy p is also linear in the order flow $x+u$. Proofs for the analytical results are provided in the Appendix.

3.1 Trading Strategy

Conjecture and later verify that the manager's equilibrium trading strategy is

$$x(s) = \beta s, \tag{3.1}$$

where $\beta > 0$ is an endogenous constant determined in the equilibrium. Conjecture (3.1) states that the manager buys (sells) the asset if her s is positive (negative), where the order size is proportional to her skill $|s|$; the more skilled the manager, the more aggressively she trades.[12] The scale factor β is referred to as the *trading intensity*, representing how intensively the manager exploits her informational advantage over the market maker.

3.2 Execution Price

The market maker decides on the price of the asset by extracting information about the asset's payoff from the aggregate order flow $x+u$. Anticipating the

[12]This conjecture follows from the original Kyle model where an informed trader trades based on the difference between her belief and that of the market maker, i.e., $\mathrm{E}[\delta|s]-\mathrm{E}[\delta] = s$.

trading strategy (3.1), the price in equation (2.1) can be rewritten as

$$p = \bar{\delta} + \lambda(x + u), \tag{3.2}$$

where

$$\lambda \equiv \frac{\beta\omega_s}{\beta^2\omega_s + \omega_u} \tag{3.3}$$

is referred to as "Kyle's lambda," a measure of the price impact of order flow.[13] Following the literature, we define a market to be *liquid* if λ is small.

3.3 Investors' Capital Allocation

By incorporating the fund proceeds $y = \pi + M$, investor i's optimization problem in (2.4) is

$$\max_{m_i \geq 0} \frac{m_i}{M}(1-\phi)\mathrm{E}\left[\pi|f\right] - \phi m_i. \tag{3.4}$$

Investor i solves this problem by incorporating the impact of her choice (m_i) on the total fund flow (M), while taking other investors' behavior ($\sum_{j\neq i} m_j$) as given. As she allocates a larger amount of capital, its marginal return diminishes due to the pro-rata allocation of the fund's proceeds, while the cost of capital linearly increases. This structure ensures the second-order condition of problem (3.4), and the first-order condition pins down the optimal capital allocation as follows:

$$m_i = \left(\frac{1-\phi}{\phi}\mathrm{E}[\pi|f]\sum_{j\neq i} m_j\right)^{\frac{1}{2}} - \sum_{j\neq i} m_j. \tag{3.5}$$

We focus on the symmetric equilibrium where all investors take the same action, $m_i = m$ for all $i = 1, 2, \cdots, n$. In this equilibrium, equation (3.5) reduces to

$$m = \frac{n-1}{n^2}\frac{1-\phi}{\phi}\mathrm{E}[\pi|f], \tag{3.6}$$

[13]Since the market maker believes that the manager follows (3.1), the linear filtering rule with normally distributed random variables yields $p = \mathrm{E}[\bar{\delta} + s|\beta s + u] = \bar{\delta} + \mathrm{E}[s] + \frac{\mathrm{Cov}[s, \beta s+u]}{\mathrm{Var}[\beta s+u]}(x + u - \mathrm{E}[\beta s + u]) = \bar{\delta} + \frac{\beta\omega_s}{\beta^2\omega_s+\omega_u}(x+u)$.

and the total fund flow becomes

$$M \equiv nm = \theta \mathrm{E}[\pi|f], \tag{3.7}$$

where $\theta \equiv \frac{n-1}{n}\frac{1-\phi}{\phi}$ represents the *flow-performance sensitivity*. Both the individual capital allocation and the total fund flow are linearly increasing in the expected trading profit conditional on the fund information, $\mathrm{E}[\pi|f]$. This is due to the performance-based fee structure, which makes the per-unit return from delegated asset management increasing in the trading profit. Also, equation (3.7) specifies the flow-performance sensitivity θ, which governs the reaction of the fund flow to the expected fund performance. The fund flow becomes less sensitive to the expected performance when the fee rate (ϕ) is high, as high ϕ raises the cost of delegation and discourages investment, while it becomes more sensitive when the fund has a large clientele (n), as it attracts capital from a larger number of investors.

3.4 Reputation

In determining the capital callocation in (3.6), each investor seeks to compute the expected fund profits $\mathrm{E}[\pi|f]$. This process, in turn, requires the investor to back out the realized skill level of the manager $|s|$ from fund information $f = x + e$.

Lemma 3.1. *Upon observing the fund information $f = x+e$, investors update their expectation about the trading profit as*

$$\mathrm{E}\left[\pi|f\right] = (1-\lambda\beta)\beta\mathrm{E}[s^2|f] = (1-\lambda\beta)\beta\left(\hat{s}^2 + \hat{\omega}_s\right), \tag{3.8}$$

where the posterior expectation of the (signed) realized skill is

$$\hat{s} = \mathrm{E}[s|f] = \xi(x+e), \tag{3.9}$$

with

$$\xi = \frac{\beta\omega_s}{\beta^2\omega_s + \omega_e}, \tag{3.10}$$

and the posterior of the skill variance is

$$\hat{\omega}_s = \mathrm{Var}[s|f] = \frac{\omega_s \omega_e}{\beta^2 \omega_s + \omega_e}. \tag{3.11}$$

Equation (3.8) suggests that the fund return, $\pi = (\delta - p)x$, becomes a quadratic function of s, as both the manager's trading quantity x and the profit margin, $\delta - p$, are proportional to s in expectation. Therefore, investors must update their belief about the manager's skill based on fund information. Equation (3.9) describes how investors handle this updating process: $\hat{s}$ represents the updated expectation about skill and is referred to as the *reputation* in the following discussion. According to equation (3.9), ξ measures how much investors rely on f in updating their belief and governs the sensitivity of the reputation to the fund's trading strategy x. Holding β constant, ξ is large when investors' prior is unreliable (ω_s is large) and the fund information is transparent (ω_e is small). More importantly, ξ itself also depends on the trading strategy β, as a more intensive trading strategy makes f more informative about s.

Lemma 3.1 implies that, on top of the maximizing profits, delegated asset management generates an additional incentive for the manager to control her trading strategy x, as it affects the total fund flow M by influencing the investors' expectation about fund's returns.

3.5 Manager's Optimization

By incorporating the total fund flow ([3.7] and [3.8]), the manager's problem in (2.3) is re-written as follows:

$$\max_{x} \; sx - \lambda x^2 + \theta(1 - \lambda\beta)\beta \left[\xi^2 (x^2 + \omega_e) + \hat{\omega}_s\right]. \tag{3.12}$$

The first two terms represent the fee income from the investment return and are essentially identical to the trading profit in the original Kyle (1985) model. The last term with coefficient θ arises from the fee income associated with the fund flow M. It increases with the manager's risk taking, measured by x^2, because she can potentially inflate her reputation $|\hat{s}|$ by placing a large order (either buy or sell). The reaction of the expected fund flow to the trading strategy is influenced not only by the flow-performance sensitivity θ but also by the investors' updating coefficient ξ. This is because fund opacity ($\omega_e > 0$) prevents investors from perfectly inferring the manager's trading strategy from fund information. Importantly, investors' estimate of the manager's skill and, therefore, the reaction of the fund flow to the risky investment depend on the risky trading strategy β itself: the more intensive the manager invests, the more reliable the fund information becomes for investors. This structure is captured by the dependence of ξ on β in equation (3.10) and leads to the strategic complementarity between the manager's risk taking and investors' capital allocation based on reputation.

The unique solution to the manager's optimization problem in (3.12) (given β) is represented by

$$x = F(\beta)s, \tag{3.13}$$

where

$$F(\beta) \equiv \frac{1}{2\left(\lambda - \theta(1-\lambda\beta)\beta\xi^2\right)}. \tag{3.14}$$

$F(\cdot)$ is a nonlinea function of β due to the dependence of λ and ξ on β.

3.6 Equilibrium under Opacity

The manager optimally chooses (3.13) given that the market maker and investors believe that she follows strategy (3.1). In equilibrium, their beliefs must be correct, meaning that (3.1) and (3.13) must be consistent with each other. This belief consistency condition requires that

$$\beta = F(\beta). \tag{3.15}$$

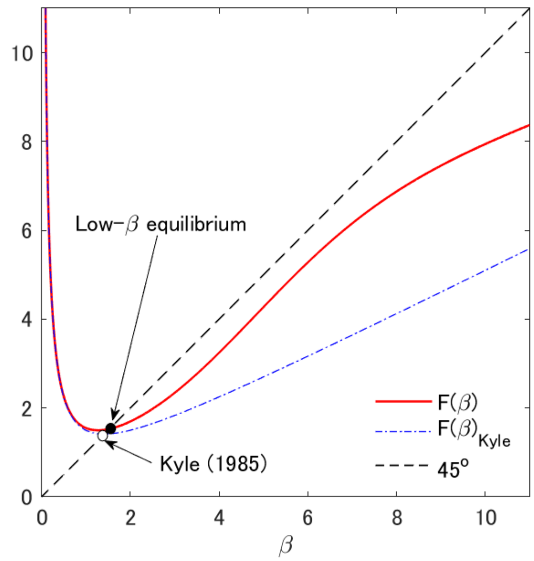
That is, the equilibrium levels of β are determined at the fixed points of $F(\cdot)$. Having identified β, the equilibrium levels of x, p, and M are determined by equations (3.1), (3.2) and (3.7), respectively. In what follows, we first illustrate the determination of equilibrium β through a numerical example, followed by an analytical proposition that formally establishes the equilibria.

Figure 2 illustrates the determination of β for three different levels of fund opacity ω_e. The intersections of $F(\beta)$ (the solid line) and the 45-degree line yield the equilibrium levels of β. We also plot $F(\beta)_{\text{Kyle}}$ in the dash-dotted line, representing the original Kyle (1985) model. This benchmark case arises when the fund is infinitely opaque (i.e., $\omega_e \to \infty$), as it makes the fund flow inelastic to the manager's trading strategy, and the delegated asset management becomes irrelevant.

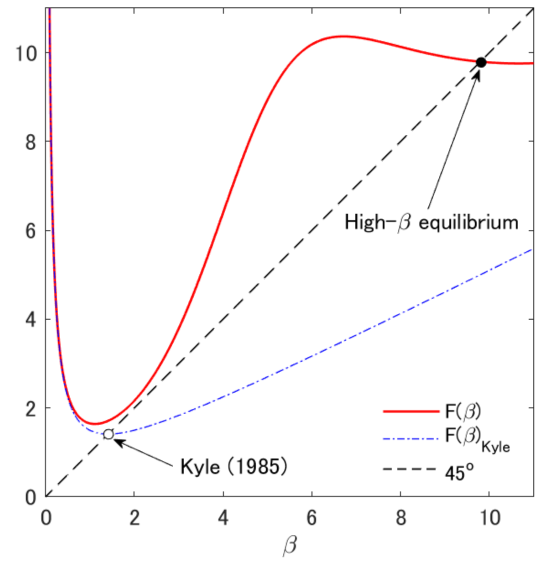
Panel (a) of Figure 2 shows that if ω_e is large, there is a unique "low-β equilibrium," in which β is close to the trading intensity without delegated asset management. Panel (b) shows that a large ω_e leads to a unique "high-β equilibrium," in which β is much larger than that in the original Kyle (1985) model. Panel (c) shows that multiple equilibria arise for an intermediate level of ω_e. Specifically, the high-β and the low-β equilibria are stable, as $F(\beta)$ crosses the 45-degree line from above, whereas the one with intermediate β is unstable, as $F(\beta)$ crosses the line from below.[14] In what follows, we focus only on the stable equilibria.

Fund opacity and multiple equilibria. To see the intuition behind the different equilibrium patterns presented in Figure 2, note that the fund manager faces two conflicting motives when trading. On the one hand, she seeks to avoid trading too aggressively on her private information, as doing so would move the price excessively and reduce her trading profits. As in Kyle (1985), this motive leads the manager to decrease β. On the other hand, she wishes to trade intensively to signal her skill to her investors, aiming to establish a strong reputation and attract large fund flows. This motive encourages her to increase β. In summary, she faces a tradeoff between "hiding" her skill from the market maker and "showing it off" to her investors. The fund opacity ω_e governs this tradeoff by influencing the sensitivity of the manager's reputation held by

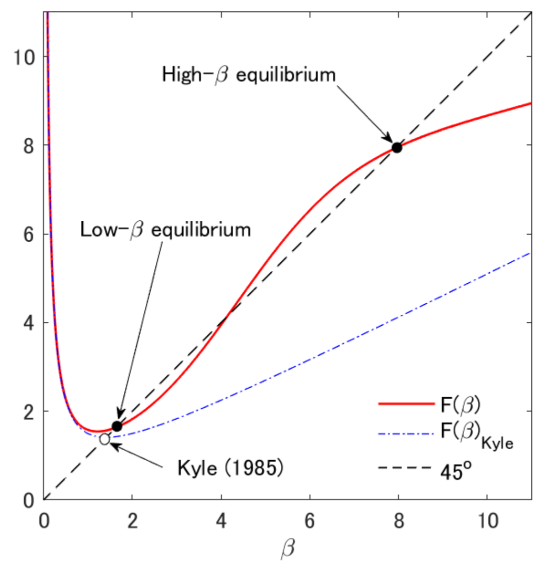
[14]We define an equilibrium to be stable (unstable) if the corresponding β is a stable (unstable) fixed point of $F(\cdot)$, that is, $|F'(\beta)| < 1$ (> 1). A stable equilibrium is robust to a small perturbation to the agents' belief about β, meaning that the economy converges back to the original equilibrium through the best-response dynamics of the agents' beliefs.



(a): Large ω_e ($\omega_e = 48$) $\Rightarrow$ unique low-β equilibrium.



(b): Small ω_e ($\omega_e = 28$) $\Rightarrow$ unique high-β equilibrium.



(c): Intermediate ω_e ($\omega_e = 38$) $\Rightarrow$ multiple high-β & low-β equilibria.

Figure 2: Determination of β

Note: The figure plots $F(\beta)$ for different values of θ. The equilibrium values of β are determined at the fixed points of $F(\cdot)$. The parameter values used in the figures are $\omega_s = 1$, $\omega_u = 2$, and $\phi = 0.02$.

investors to her investment behavior. In particular, Lemma 3.1 demonstrates that the fund flow M becomes less sensitive to x under opacity (i.e., ξ is small) compared to the fully transparent case, as it becomes difficult for investors to estimate the skill based on the fund information f. Therefore, fund opacity diminishes the strategic complementarity between the manager's risk taking and the reputation-driven capital allocation by investors, thereby weakening the showing-off motive.

For an opaque fund with a large ω_e (panel [a]), the hiding motive dominates the showing-off motive, leading to a unique low-β equilibrium. In this equilibrium, investors believe that the manager's trading intensity is relatively low and the fund information is not very informative, making the reputation and the fund flow insensitive to x. Although the manager could inflate her reputation by placing an order larger than what her investors anticipate, she refrains from doing so, because it would cause a large price impact and reduce the fund's investment profits, while fund flows and her fee income would not increase much due to small ξ. Consequently, the equilibrium β is close to the trading intensity without delegated asset management, meaning that it is a "Kyle-like equilibrium."

For a transparent fund with a small ω_e (panel [b]), the showing-off motive prevails, resulting in a unique high-β equilibrium. In this equilibrium, investors believe that the manager is trading aggressively. The reputation and fund flows grow sensitive to the manager's trading strategy through the fund information. Under such beliefs, the manager is actually willing to trade aggressively: although it moves the price substantially and undermines investment profits, it helps her build a strong reputation and generates a large fee income from significant capital inflows. Importantly, investors' learning is not manipulated, as their belief about the manager's trading strategy is correct. Nonetheless, the manager places a large order because doing otherwise would induce her investors to incorrectly estimate that she is less skilled, thereby deteriorating her reputation. She trades aggressively solely to conform to her investors' belief and to attract large fund flows. To emphasize this mechanism, we call this equilibrium the "flow-driven equilibrium."

For intermediate levels of ω_e (panel [c]), the strategic complementarity between investors' reputation-based capital allocation and the manager's risk taking is neither weak nor strong, and there is no clear dominance between the two motives. Consequently, both the low-β and the high-β equilibria are supported under intermediate fund opacity. Each equilibrium is associated with a self-fulfilling belief of the market maker and investors.[15]

The following proposition summarizes the above discussion and provides the analytical characterization of the equilibria.

Proposition 3.1 (Equilibrium characterization)**.** *There exists a critical value of the flow-performance sensitivity, denoted as θ_0.*

(i) If $\theta \leq \theta_0$, the equilibrium is unique and characterized by a threshold of fund opacity ω_0, such that, $\omega_e \leq \omega_0$ leads to the high-β equilibrium and $\omega_e \geq \omega_0$ leads to the low-β equilibrium.

(ii) If $\theta > \theta_0$, there are two thresholds of fund opacity, ω_L and $\omega_H (> \omega_L)$. When $\omega_e \leq \omega_L$, the high-β equilibrium is unique; when $\omega_e \geq \omega_H$, the low-β equilibrium is unique; and when $\omega_e \in (\omega_L, \omega_H)$, the high- and low-$\beta$ equilibria coexist.

Figure 3 illustrates the result in Proposition 3.1. It indicates that both fund opacity ω_e and flow-performance senitivity θ are critical in determining the equilibrium type, as they jointly influence the reaction of the fund flow to changes in the manager's trading strategy, thereby affecting her showing-off motive. As high levels of θ strengthen this motive, we obtain the following corollary:

Corollary 3.1. *(i) When $\theta \leq \theta_0$, the threshold of fund opacity ω_0 is monotonically increasing in θ with $\lim_{\theta \to 0} \omega_0 = 0$.*

(ii) When $\theta > \theta_0$, both ω_L and ω_H are increasing in θ. Moreover, the region for multiple equilibria, (ω_L, ω_H), expands as θ increases.

[15] For simplicity, we assume that agents do not agree to disagree, so that they coordinate their belief when parameter values are such that multiple equilibria exist.

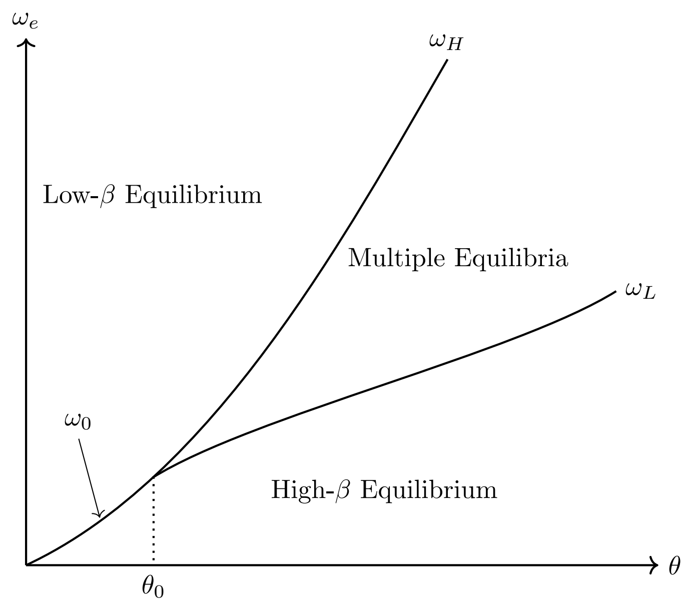


Figure 3: Equilibrium Characterization

Note: This figure illustrates the equilibrium states using the flow-performance sensitivity (θ) and fund opacity (ω_e). When $\theta \leq \theta_0$, the cutoffs of opacity converge to $\omega_H = \omega_L = \omega_0$.

For a given value of fund opacity ω_e, a weak flow-performance sensitivity supports only the low-β equilibrium, as the hiding motive strictly dominates the weak showing-off motive. However, as θ increases, a high flow-performance sensitivity encourages the manager to show off her skill by trading intensively. Hence, it gives birth to the high-β equilibrium. The tradeoff between the hiding and the showing-off motive tends to balance and generates multiple equilibria when the fund is moderately opaque and fund flows are sensitive to the expected investment performance. Otherwise, one of the conflicting motives dominates the other, making either the Kyle-like equilibrium or the flow-driven equilibrium the unique equilibrium.

Equilibrium comparison. Comparing the two stable equilibria, does the manager perform better in one of them? What are the implications for market liquidity, asset prices, and learning about asset-selection skills?

Corollary 3.2 (High-β vs. low-β equilibrium)**.** *In the high-β equilibrium, relative to the low-β equilibrium, the following results hold.*

(i) The market is more liquid, that is, λ is lower;

(ii) The manager's expected investment performance given the realized skill, $\mathrm{E}[(\delta - p)x|s] = \frac{\beta\omega_u s^2}{\beta^2\omega_s+\omega_u}$, is lower;

(iii) The price informativeness, defined by $\tau \equiv \frac{1}{\mathrm{Var}[\delta|p]} = \frac{\beta^2\omega_s+\omega_u}{\omega_\delta(\beta^2\omega_s+\omega_u)+\omega_u\omega_s}$, is higher;

(iv) The price volatility, $\mathrm{Var}[p] = \frac{\beta^2\omega_s^2}{\beta^2\omega_s+\omega_u}$, is higher; and

(v) Investors' estimate of the manager's skill is more precise, that is, $\hat{\omega}_s = \mathrm{Var}[s|f] = \frac{\omega_s\omega_u\omega_\delta}{\omega_s(\beta^2\omega_\delta+\omega_u)+\omega_u\omega_\delta}$ is lower.

Statement (i) of Corollary 3.2 holds because price impact λ is decreasing in β in the equilibrium. Namely, the information is exaggerated in that the order size is too large relative to the true $|s|$ due to the showing-off motive. Thus, the market maker attempts to partially "undo" the manager's information revelation by incorporating less of it into the price, resulting in a weaker price impact. Since the high-β equilibrium emerges only if the fund is relatively transparent, the statement also implies that a higher fund transparency leads to higher market liquidity.

The intuition for statement (ii) is closely related to that for statement (i). In the high-β equilibrium, the manager earns a lower trading profit due to a smaller informational advantage over the market maker. This result may appear inconsistent with statement (i) because, ceteris paribus, a lower price impact is typically associated with a higher trading profit. However, the manager's order size $|x|$ in the high-β equilibrium is so large that the *overall* price reaction (i.e., λ multiplied by $|x|$) is larger than that in the low-β equilibrium. Consequently, the manager moves the price excessively and earns a lower trading profit in this equilibrium. Due to her aggressive trading, more information about s, which is informative about δ, is incorporated into the price in the high-β equilibrium, supporting statement (iii).

Statement (iv) follows, again, from the manager's aggressive trading in the high-β equilibrium. Specifically, she places a large buy (sell) order when $s > 0$ ($s < 0$), even if $|s|$ is small. This causes a large upward (downward) pressure on p, leading to greater price variability. This result aligns with the insight of Guerrieri and Kondor (2012) that reputation concerns amplify price volatility, although their mechanism differs from ours.[16]

In the high-β equilibrium, the manager reveals more information about s, thereby improving the precision of investors' estimation, as statement (v) demonstrates. This makes fund flows more responsive to skill, as investors can assess it more accurately.

3.7 Implications

This section discusses economic implications drawn from the equilibrium analyses under opacity.

[15] λ represents the price's reaction to the order flow $x + u$, rather than its reaction to the manager's private information s. The price impounds s with coefficient $\beta\lambda$, which is monotonically increasing in β.

[16] Guerrieri and Kondor (2012) argue that bond price volatility increases because fund managers demand a premium to offset the risk of reputation damage in the event of default.

3.7.1 Financial Fragility

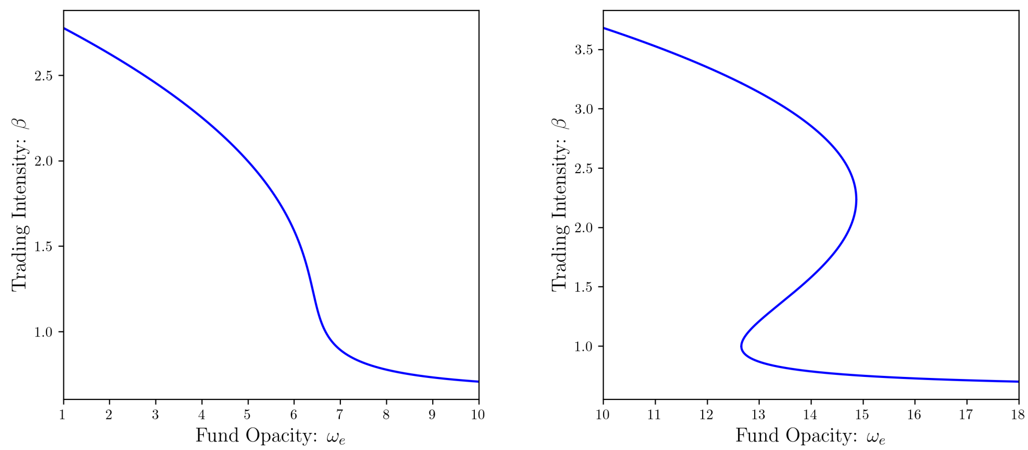
The self-fulfilling nature of multiple equilibria suggests that financial markets may become vulnerable to non-fundamental factors. A shift in investor beliefs alone, without any structural changes, could cause the market to transition from the standard Kyle-like equilibrium to the flow-driven equilibrium, characterized by high price volatility and poor trading performance. We interpret this phenomenon as a form of *fragility* in financial markets.

Our model suggests that fragility may occur when flow-performance sensitivity (θ) is sufficiently high. This result is consistent with empirical studies documenting that strong flow-performance sensitivity can amplify trading pressure and contribute to market fragility by inducing managers to adjust trading behavior in response to investor flows rather than fundamentals (e.g., Coval and Stafford, 2007; Goldstein, Jiang, and Ng, 2017). While this literature typically attributes fragility to flow dynamics without explicitly focusing on the size of the investor base, our results suggest that a larger clientele (n) mechanically increases θ and thereby sows the seeds of market fragility.

A high flow-performance sensitivity can also arise when the fund management fee (ϕ) is relatively low. This surprising result goes counter to the recent debate on performance-based compesations for fund managers and advisory contracts, where payment structures that strongly reward performance are often criticized for encouraging excessive risk-taking. In our model, such performance-based compensation can actually stabilize the market by eliminating the flow-driven equilibrium and guiding the economy toward a unique low-β equilibrium. This is consistent with the empirical finding by Dass, Massa, and Patgiri (2008) that high-powered advisory contracts do not induce investment in bubbly stocks but instead reduce such investments, highlighting the stabilizing effect of incentive compensations on financial markets.

3.7.2 Opacity and Trading Style

Furthermore, Proposition 3.1 yields novel implications connecting the fund opacity and its trading styles.



(A) Low Flow-performance Sensitivity

(B) High Flow-performance Sensitivity

Figure 4: Fund Opacity and Trading Intensity

Note: These figures are illustrated by seeting $\omega_s = 1.0, \omega_u = 2.0, \theta = 10 (\phi = 0.08)$ for the left panel and $\theta = 27 (\phi = 0.02)$ for the right panel.

Corollary 3.3. *Both in the high- and low-β equilibria, the trading intensity β is monotonically decreasing in the fund opacity ω_e.*

Figure 4 illustrates the result in Corollary 3.3: the left panel represents the case with a unique equilibrium by setting a weak flow-performance sensitivity, while the right panel involves multiple equilibria due to a strong flow-performance sensitivity. First, as both panels illustrate, opaque funds tend to adopt low-intensity strategies (i.e., the low-β equilibrium) that minimize market impact, prioritizing profits over fund flows. In contrast, transparent funds engage in high-intensity strategies with flashy risk taking (i.e., the high-β equilibrium), seeking reputation among investors and large fund flows. This relationship arises because the manager's reputation becomes insensitive to her trading behavior when the fund information is opaque and difficult to interpret for fund investors. The model predicts that these tendencies are more pronounced when the flow-performance sensitivity is weak, as the equilibrium is unique regardless of fund opacity (panel [a] of Figure 4).

These results are consistent with common observations in the real market. For example, quantitative hedge funds are often extremely opaque and highly profit-driven, focusing on consistent performance, with benefits from attracting fund flows being regarded as secondary to profitability (Schwager, 2012). On the other hand, some transparent funds, such as certain thematic ETFs or crypto-focused funds, engage in bold, attention-grabbing strategies to attract investor flows. These funds often emphasize visibility and narrative-driven positioning, which can lead to volatile returns and greater risk taking.

The second implication is drawn from the model's multiple equilibria, as panel (b) in Figure 4 illustrates. Namely, the relation between funds' activities (β) and their transparency (ω_e) may become blurry when the opacity is at an intermediate level. Notably, this indeterminacy may lead to financial fragility. Depending on market beliefs, relatively opaque funds may shift between profit-seeking and flow-driven strategies, creating non-fundamental fluctuations. Our result suggests that even relatively opaque funds may follow the crowd due to market beliefs, hopping between safe and risky strategies.

A notable observation in line with this result is momentum crashes (Daniel and Moskowitz, 2016), where funds engaging in risky and visible momentum trades often switch to safer strategies, generating sharp reversals and amplifying price declines. More generally, the rapid shifts between risk-taking and solid performance-based behavior have contributed to market fragility and volatility, as documented in various studies on economic crises, such as Brunnermeier (2009) on the Global Financial Crisis in 2007. Importantly, large fundamental shifts that support such transitions are rarely identified (Gennotte and Leland, 1999). Our model does not require such fundamental shocks, as a change in beliefs alone can trigger a shift.

3.7.3 Compensation in Fund Management Industry

Our model provides insights into how skills are reflected in compensation in the fund management industry. As shown in statement (v) of Corollary 3.2, fund investors learn s more precisely in the high-β equilibrium than in the low-β equilibrium, as the manager demonstrates her skill by trading more intensively. Consequently, in the high-β equilibrium, the manager's expected compensation, $\mathrm{E}[\phi y|s]$, become more sensitive to her realized skill. As an

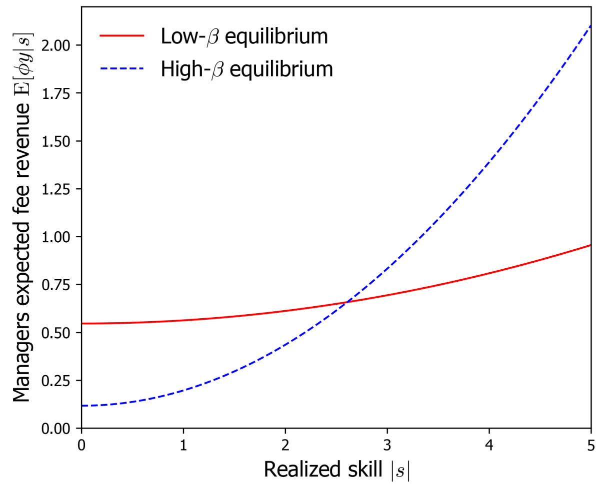


Figure 5: Skill versus Expected Wage

Note: The figure plots the manager's expected fee income at the low-β equilibrium (solid line) and high-β equilibrium (dashed line) for different levels of skill, $|s|$. The parameter values used in the figures are $\omega_s = 1$, $\omega_u = 2$, $\omega_e = 38$, and $\theta = 27$.

illustration, Figure 5 plots the manager's expected compensation in relation to her skill in the low- and high-β equilibria. In the high-β equilibrium, investors' accurate evaluations result in greater pay inequality: low-skilled managers would receive very low compensation, while high-skilled ones are very highly paid. This result corroborates empirical observations, such as those provided by Philippon and Reshef (2012) and Böhm, Metzger, and Strömberg (2018), highlighting the comparatively large wage inequality in the fund management industry. Célérier and Vallée (2019) attribute its source to the scalability of skill, while inequality in our model arises from managers' incentives to attract fund flows through establishing high reputation.

3.7.4 Skill and Performance Persistence

Do high-skilled managers consistently perform well? Not always: skilled managers may not necessarily outperform less skilled ones. Consider a highly skilled manager and a relatively less skilled one. Since trading profits are increasing in managers' skill $|s|$, the former is expected to achieve higher trading

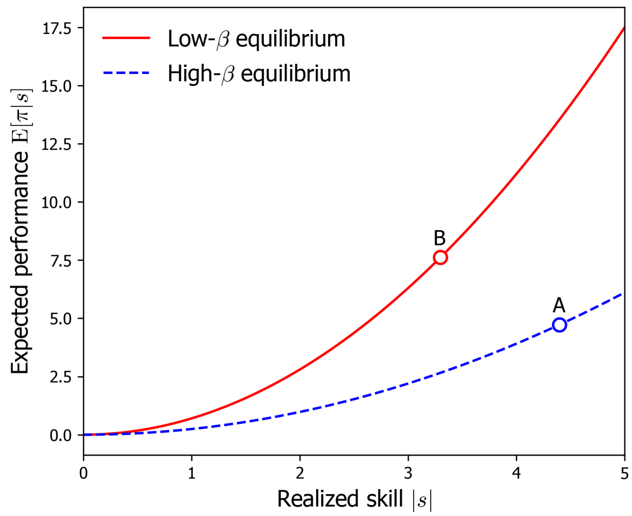


Figure 6: Skill versus Expected Trading Profit

Note: The figure plots the expected trading profit $\mathrm{E}[\pi]$ at the low-β equilibrium (solid line) and the high-β equilibrium (dashed line) for different levels of skill $|s|$. The parameter values used in the figure are $\omega_s = 1$, $\omega_u = 2$, $\omega_e = 38$, and $\phi = 0.02$.

profits than the latter *conditional on* both managers trading at the same intensity β. However, if their investors believe, for whatever reason, that the skilled manager invest more aggressively than the less skilled one, and managers conform to such beliefs, a high skill level does not translate into performance. This is because the skilled manager, trading at the flow-driven equilibrium, reveals too much private information to the market maker. Hence, compared to the less skilled manager, who plays the Kyle-like equilibrium, the skilled manager may earn lower expected profits. Figure 6 illustrates such a situation.[17] It plots the expected trading profits (performance) at high- and low-β equilibria with different levels of skill, suggesting that a low-skilled manager in the low-β equilibrium (point B) outperforms a high-skill manager in the high-β equilibrium (point A).

The empirical literature (e.g., Jensen, 1967; Busse, Goyal, and Wahal, 2010) finds that asset managers do not outperform passive benchmarks on

[17]The same argument holds for the comparison of the unconditional expected profit, $\mathrm{E}[\pi]$, using skill variance ω_s as an *ex-ante* measure of the skill level.

average and, in cross-section, those who outperform cannot do so consistently. While the literature attributes the lack of persistent performance to the lack of skill, our result proposes an alternative explanation. Several studies reconcile the existence of skill with the lack of persistent performance. Berk and Green (2004), for example, argue that skill-based alphas eventually disappear due to competition between investors and decreasing returns to scale at the fund level.[18] Our model contributes to the understanding of this matter by showing that the lack of interplay between manager performance and skill could be the result of self-fulfilling multiple equilibria.

4 Strategic Opacity

This section considers endogenous opacity by allowing the manager to control ω_e. It uncovers the manager's incentive to obfuscate fund information and analyzes how it leads to the financial fragility.

4.1 Setup

Obfuscating fund information. Prior to investors' capital allocation and the manager's investment decision, the fund manager chooses the fund opacity ω_e, anticipating the equilibrium reactions of investors and the market maker specified in Section 3. We interpret this behavior as strategic obfuscation by the fund manager. For example, in many mutual funds, managers provide information about their strategies to investors in accordance with regulatory requirements. However, they strategically design the way this information is presented and convey a deliberately uninformative description even when the underlying holdings are simple. We assume that the manager pays the obfuscation cost, $C(\omega_e)$ with $C'(\cdot) > 0$ and $C(0) = 0$, to increase the degree of opacity of fund information. It could be thought of as pecuniary costs arising from investments in communication and marketing resources to design

[18] Pastor, Stambaugh, and Taylor (2015) document increases in skill in finance and attribute the lack of corresponding increases in performance to decreasing returns to scale at the industry level.

abstract narratives and vague labels that replace direct descriptions of risky portfolios. It may also reflect non-pecuniary effort that managers incur to carefully structuring disclosure documents to maintain opacity while following regulatory compliance. In what follows, we assume that the marginal cost of opacity $C'(\cdot)$ is sufficiently large to ensure the second-order condition of the manager's problem defined below.

Sunspot shock. To formalize the manager's optimization problem in the rational expectations framework, we introduce an equilibrium-selection device in cases of multiple equilibria. In particular, following the most common approach, we assume that market participants condition their actions on the realization of an exogenous *sunspot* shock, characterized by the binary random variable $z \in \{0, 1\}$.[19] Namely, whenever parameters support multiple equilibria, market participants play the high-β equilibrium if a spot appears on the sun ($z = 1$), whereas the low-β equilibrium is realized if no spots are observed ($z = 0$). Accordingly, the high- and low-β equilibria are realized with probability $\rho_H \equiv \Pr(z = 1)$ and $\rho_L = 1 - \rho_H$, respectively. The sunspot shotck is realized after the fund manager sets fund opacity but prior to the fund-raising stage.

Manager's expected utility. Incorporating the cost, the manager controls ω_e to maximize her *ex-ante* expected utility derived from the fee income, $\mathrm{E}[\phi y]$. As the fund flow is given by equation (3.7), her objective function is specified as follows.[20]

$$U = \phi \mathrm{E}[\pi + M] = \phi(1 + \theta)\mathrm{E}[\pi] - C(\omega_e), \tag{4.1}$$

where the expectation operator incorporates the sunspot shock together with all other random variables introduced so far.

[19]The sunspot equilibrium is proposed by Diamond and Dybvig (1983) as an equilibrium selection device in the context of bunk runs and formalized in the equilibrium analyses by the subsequent studies, such as Cooper and Ross (1998).

[20]The law of iterated expectation implies $\mathrm{E}[M] = \theta \mathrm{E}[\mathrm{E}[\pi|f]] = \theta \mathrm{E}[\pi]$.

Equilibrium. In this extended model, the definition of equilibrium augments Definition 1 of the baseline model by additionally requiring that ω_e maximizes the manager's objective function U in equation (4.1).

4.2 Opacity as Commitment Device

Before we solve problem (4.1), consider the manager's incentive to obfuscate fund information. Given the trading intensity β, the unconditional expectation of fund performance in (4.1) is computed as

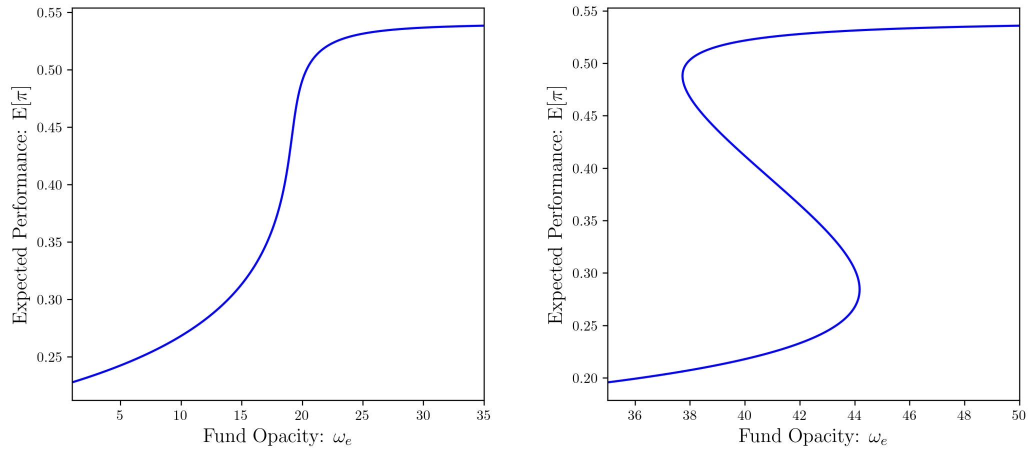
$$\mathrm{E}[\pi] = \frac{\beta\omega_u}{\beta^2\omega_u + \omega_s}\omega_s. \tag{4.2}$$

As in the standard Kyle (1985) framework, the first fraction represents the profit margin that each unit of informational advantage would earn, and the second term ω_s represents the expected size of this advantage.

Fund opacity ω_e does not directly show up in equation (4.2), but it influences the expected performance through the equilibrium trading intensity β.

Corollary 4.1. *Both in the low- and high-β equilibria, the expected fund performance* $\mathrm{E}[\pi]$ *is monotonically increasing in fund opacity* ω_e.

Figure 7 plots $\mathrm{E}[\pi]$ in relation to ω_e, which traces the reactions of β in Figure 4. In our model, $\mathrm{E}[\pi]$ declines with the trading intensity β. This arises from the presence of the showing off motive. Although unconditional expected performance is maximized at the Kyle benchmark, which is driven solely by the hiding motive, the manager in our model trades more aggressively in order to attract fund flows. It revelas too much private information to the market maker and lowers the fund performance, while the manager is willing to deviate from the profit-maximizing strategy because the fund flow compensates for this loss. The opacity, conversely, diminishes the showing-off motive and induces the manager to take more profit-driven strategies, bringing β closer to the profit-maximizing level. As a consequence, opacity helps improve the



(A) Low Flow-performance Sensitivity (B) High Flow-performance Sensitivity

Figure 7: Fund Opacity and Expected Performance

Note: The figure plots the unconditional expected performance of the fund $\mathrm{E}[\pi]$ against fund opacity ω_e with different values of the flow-performance sensitivity: $\theta = 10$ (the left panel) and $\theta = 27$ (the right panel). The value of other parameters are $\omega_s = 1.5$ and $\omega_u = 3$.

expected fund performance and, according to the manager's utility (4.1), this effect encourages the manager to obfuscate fund information.

Interestingly, investors' *ex-ante* expected utility is also increasing in fund opacity. According to equation (3.6), each investor expects to obtain

$$U_i \equiv \frac{1-\phi(1+\theta)}{n}\mathrm{E}[\pi], \tag{4.3}$$

so that her expected utility is proportional to the fund's unconditional expected performance. Thus, Corollary 4.1 implies that even investors prefer opaque fund from the *ex-ante* perspective. This result may appear paradoxical in light of the showing-off motive: why does the manager trade aggressively and deviate from the performance-maximizing strategy (the Kyle benchmark), despite knowing that such behavior lowers performance and reduces both her own and investors' expected utility? This tension arises from a crucial information timing gap. When she decides on the trading strategy x, the manager cannot alter the market's beliefs about her strategy (β and λ in equations [3.8]–[3.11]), meaning that she optimizes over x by taking the investors' updating rule as given. She can influence investor perceptions only through the content of fund information f by inflating her trading volume $|x|$ to appear more skilled, creating the showing-off motive. Conversely, at the *ex-ante* stage (i.e., before she sends the fund information to investors), the manager recognizes that this future showing-off motive will lead to flashy but inefficient risk-taking that erodes unconditional performance and the expected fund flow $\mathrm{E}[M]$. Consequently, she strategically chooses fund opacity as a commitment device to "tie her own hands," thereby dampening the investors' future responsiveness to f and forcing her future self to prioritize profit-driven trading over flow-driven trading.

4.3 Equilibrium Opacity

1. Analytically show the condition such that optimal opacity falls in the region of multiple equilibria, i.e., $\omega_e^* \in (\omega_L, \omega_H)$.
2. The optimal opacity is either constant or increasing in relation to the expected skill.

4.3.1 Skill, Opacity, and Fragility

We first propose "middling" skill acquisition due to the multiple trading-stage equilibria by focusing on $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$. We refer to it as middling because the acquired skill level is either higher or lower than what she would acquire if she knew which type of equilibrium would be realized. On the equilibrium path, both the selected skill potential (ω_s^*) and the market's belief about it (ω_s) fall in this region if the marginal skill-acquisition cost is intermediate. In this region, the trading stage involves multiple equilibria (i.e., $\beta_0 = \beta_H$ or β_L), whereas the trader at the skill-acquisition stage cannot influence the equilibrium selection. Consequently, she determines her skill potential by anticipating the sunspot shock, which leads to the skill acquisition as follows.

Proposition 4.1 (Middling skill acquisition)**.** *The equilibrium skill potential in the presence of multiple trading-stage equilibria is* *(i) lower than the skill potential that she would acquire in the low-β equilibrium; and*

(ii) higher than the skill potential that she would acquire in the high-β equilibrium.

Figure **??** illustrates a numerical example; $\omega_{s,m}$ is the ex-ante optimal skill potential, while the ex-post equilibrium is either $\omega_{s,\ell}$ or $\omega_{s,h}$, depending on the equilibrium realization in the trading stage.

The trader invests in skill potential incorporating the possibility of both high- and low-β equilibria. At the trading stage, however, the acquired skill potential turns out to be insufficient if the low-β equilibrium is realized, as its optimal play involves the profit-driven strategy and requires a relatively high asset-selection skill (i.e., $\omega_{s,h}$ is the *ex-post* equilibrium). Conversely, the acquired skill potential is too high if the high-β equilibrium is realized, as the trader takes the reputation-driven strategy that does not require a high skill potential (i.e., $\omega_{s,\ell}$ is the *ex-post* equilibrium).

This result indicates that individuals or firms may invest in skills that seem oddly intermediate or cautious—neither fully specialized nor overly basic. For example, it explains why some traders and finance professionals choose to pursue broader business qualifications instead of more specialized academic paths, allowing them to adapt to a wide range of market situations. Traders also learn intermediate-level coding (e.g., Python) for algorithmic trading or data analysis but stop short of deep computer science expertise. A similar trend can be observed with respect to CFA Levels and quantitative skills among hedge fund managers. These skill acquisitions would be difficult to rationalize in an economy with a single equilibrium but become logical under the equilibrium multiplicity and uncertainty regarding equilibrium realization.

4.3.2 Fragile Skill Acquisition and Skill Diversity

By expanding our scope to the entire range of skill potential (ω_s), the discontinuities in Lemma **??** imply that the skill-acquisition problem in (**??**) may have multiple solutions.[21]

For example, when the marginal cost of skill acquisition is moderately small, as Figure **??** illustrates, both the high skill potential ($\omega_{s,h}$) and the intermediate skill potential ($\omega_{s,m}$) become optimal for the trader. Corresponding to $\omega_{s,h}$, the equilibrium in the trading stage is unique with $\beta_0 = \beta_L$, where the trader engages in solid profit-driven trading. Conversely, the trading stage with skill potential $\omega_{s,m}$ involves multiple equilibria, and the selected skill potential is middling, as Proposition 4.1 demonstrates. The symmetric argument holds when the marginal cost is relatively high: the intermediate skill potential and the low skill potential show up as multiple optima. This result can explain at least three realistic phenomena without relying on ad-hoc assumptions on the skill-acquisition environment.

Firstly, the coexistence of traders with different skill levels arises not from heterogeneity in skill-acquisition problems (e.g., different marginal costs), but from the multiple trading-stage equilibria and the possibility of mixed strategies in the skill acquisition stage. This reflects a realistic scenario where, due to advancements in the internet and social media (e.g., online webinars via Zoom and academic courses uploaded on YouTube), opportunities for skill acquisition have become widely and uniformly accessible to all traders, reducing barriers to learning. Despite the increased homogeneity in skill-acquisition costs, we continue to observe heterogeneity in skill levels in the finance industry. Our model offers an explanation for this observation, showing how diverse skill levels can emerge among market participants, without the need to assume inherent differences in their skill acquisition costs.

Secondly, it also captures the possibility of widespread surges in skill acquisition without assuming special conditions for the skill-acquisition environment. For example, we can explain traders and fund managers in competitive environments (e.g., Wall Street) may collectively strive to acquire high-level skills, such as pursuing advanced degrees, driven by the mixed-strategy equi-

[21] Although the equilibrium multiplicity in the trading stage (i.e., β_0) is determined by the level of ω_s, as Proposition **??** stipulates, it is orthogonal to the existence of multiple solutions to the optimal skill acquisition in equation (**??**).

librium. This surge in skill acquisition could occur even in the absence of significant changes in external factors, and without the need for heterogeneity in skill-acquisition costs, reflecting industry-wide trends in upskilling and degree inflation, as documented by Philippon and Reshef (2012) and Fuller and Raman (2017).

Finally, discontinuities in the marginal benefit curve add another layer of insight into the effect of the skill-acquisition cost on the equilibrium skill potential. Namely, discontinuities introduce the possibility of significant jumps in the optimal skill level triggered by a small change in the marginal skill-acquisition cost. For instance, when marginal technological advancements slightly reduce the marginal cost of learning, the result could be a sizable upward shift in skill acquisition. Conversely, even a small increase in the cost of learning might cause a large decline in the equilibrium skill level. This discontinuity-driven sensitivity can help explain the historical observations stylized by Philippon and Reshef (2012) that certain external changes lead to abrupt shifts in the distribution of skill levels in the finance industry.

5 Conclusion

This paper studies a model à la Kyle (1985) where traders are privately informed about their skill, i.e., the trader-specific ability to identify potentially mispriced assets. The traders care not only about trading profits but also the recruiter's perception of their skill. The model has two stable self-fulfilling equilibria, which we interpret as fragility in financial markets. One of the equilibria is similar to that of Kyle (1985) where reputation is irrelevant. In the other, "reputation-driven" equilibrium, the traders trade more aggressively for fear of being seen as unskilled by the recruiter, causing higher price volatility, higher market liquidity, and lower trading profits. Our model predicts that fragility may arise when poaching is (moderately) active in the finance industry and when there are high demands for financial skill. The model yields implications connecting traders' skills, their trading styles, and financial fragility. Moreover, consistent with empirical observations, our model predicts large pay inequality in the finance industry and reconciles skill in active management and the lack of performance persistence.

As a direction for future research, it would be interesting to analyze various types of wage structures in financial markets, ranging from back-end to front-end compensations. The implications of the forward-looking trading motives introduced in our model can be applied to investigate which types of wage structures and employment systems can mitigate financial market fragility while also improving the welfare of market participants.

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Online Appendix

A Proof of Lemma ??

From (3.1) and (3.2), p is observationally equivalent to $\frac{p-\hat{\delta}}{\lambda} = \beta s + u \equiv z_p$, which is the sum of two normally distributed variables from the market maker's perspective. Since $\delta \sim N(\hat{\delta} + s, \omega_\delta)$ is equivalently presented as $\delta = \hat{\delta} + s + \epsilon$ with $\epsilon \sim N(0, \omega_\delta)$, δ is observationally equivalent to $\delta - \hat{\delta} = s + \epsilon \equiv z_\delta$, which is also the sum of normal variables. Thus, given $Z = [z_p,\ z_\delta]^{\mathrm{T}}$, the market maker's posterior mean of s is

$$\mathrm{E}\left[s|Z\right] = \underbrace{\mathrm{E}[s]}_{=0} + \Sigma_{sZ}\Sigma_Z^{-1}(Z - \underbrace{E[Z]}_{=[0\ \ 0]^{\mathrm{T}}}), \tag{A.1}$$

where

$$\Sigma_{sZ} = \begin{bmatrix} \beta\omega_s, & \omega_s \end{bmatrix} \text{ and } \Sigma_Z = \begin{bmatrix} \beta^2\omega_s + \omega_u, & \beta\omega_s \\ \beta\omega_s, & \omega_s + \omega_\delta \end{bmatrix}. \tag{A.2}$$

Since

$$\begin{aligned} \det[\Sigma_Z] &= (\beta^2\omega_s + \omega_u)(\omega_s + \omega_\delta) - \beta^2\omega_s^2 \\ &= \omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta, \end{aligned} \tag{A.3}$$

it holds that

$$\begin{aligned} \mathrm{E}\left[s|Z\right] &= \begin{bmatrix} \beta\omega_s, & \omega_s \end{bmatrix} \frac{1}{\det[\Sigma_Z]} \begin{bmatrix} \omega_s + \omega_\delta, & -\beta\omega_s \\ -\beta\omega_s, & \beta^2\omega_s + \omega_u \end{bmatrix} \begin{bmatrix} z_p \\ z_\delta \end{bmatrix} \\ &= \frac{\omega_s}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} \left(\beta\omega_\delta z_p + \omega_u z_\delta\right) \\ &= \frac{\omega_s(\beta\omega_\delta + \omega_u)}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} \left(\frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u} z_p + \left(1 - \frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u}\right) z_\delta \right) \\ &= \xi\left(\gamma z_p + (1-\gamma) z_\delta\right), \end{aligned} \tag{A.4}$$

where

$$\xi \equiv \frac{\omega_s(\beta\omega_\delta + \omega_u)}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} \tag{A.5}$$

and

$$\gamma \equiv \frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u}. \tag{A.6}$$

Re-writing z_p and z_δ in (A.4) in terms of p and δ, we obtain the required result. $\square$

B Proof of Lemma ??

Let us present $\delta \sim N(\hat{\delta} + s, \omega_\delta)$ equivalently as $\delta = \hat{\delta} + s + \epsilon$ with $\epsilon \sim N(0, \omega_\delta)$. Then, squaring (??), we have

$$\begin{aligned} \hat{s}^2 &= \xi^2 \left(\gamma^2 x^2 + 2\gamma x \left(\gamma u + (1-\gamma)(\delta - \hat{\delta}) \right) + \left(\gamma u + (1-\gamma)(\delta - \hat{\delta}) \right)^2 \right) \\ &= \xi^2 \left(\begin{array}{c} \gamma^2 x^2 + 2\gamma^2 x u + 2\gamma(1-\gamma)(s+\epsilon)x \\ +\gamma^2 u^2 + 2\gamma(1-\gamma)(s+\epsilon)u + (1-\gamma)^2(s+\epsilon)^2 \end{array} \right). \end{aligned} \tag{B.1}$$

Taking the expectation of (B.1) from the trader's perspective,

$$\mathrm{E}[\hat{s}^2|s] = \xi^2 \left(\gamma^2 x^2 + 2\gamma(1-\gamma)sx + A \right), \tag{B.2}$$

where

$$A \equiv \gamma^2\omega_u + (1-\gamma)^2(s^2 + \omega_\delta) \tag{B.3}$$

is the term that the trader is unable to influence by her choice of x. We can omit A in the maximization problem below. Using (B.2), the trader's problem (??) is rewritten as

$$\begin{aligned} &\max_x \ \mathrm{E}\left[(\delta - p)x|s\right] + \theta\mathrm{E}[\hat{s}^2|s] \\ \Rightarrow \ &\max_x \ \mathrm{E}\left[\left(\hat{\delta} + s + \epsilon - \left(\hat{\delta} + \lambda(x+u) \right) \right) x \middle| s \right] + \theta\mathrm{E}[\hat{s}^2|s] \\ \Rightarrow &\max_x \ sx - \lambda x^2 + \theta\xi^2 \left(\gamma^2 x^2 + 2\gamma(1-\gamma)sx \right). \quad \square \end{aligned} \tag{B.4}$$

C Proof of Lemma ??

At the beginning of $t = 1$, the trader does not have reputation concerns anymore. Her problem is just to choose x_1 to maximize the expected trading profit, that is,

$$\max_{x_1 \in (-\infty,\infty)} \mathrm{E}[(\delta_1 - p_1)x_1|s]. \tag{C.1}$$

We conjecture and later verify that the trader's equilibrium order in $t = 1$ is

$$x_1 = \beta_1 s, \tag{C.2}$$

where $\beta_1 > 0$ is a constant to be determined. Given (C.2), the market maker sets the price

$$\begin{aligned} p_1 =& \mathrm{E}\left[\delta_1 | x_1 + u_1\right] \\ =& \mathrm{E}[\mathrm{E}[\delta_1|x_1+u_1, s]|x_1+u_1] \\ =& \mathrm{E}[\hat{\delta}_1 + s|x_1+u_1] \\ =& \hat{\delta}_1 + \frac{\mathrm{Cov}[s, x_1+u_1]}{\mathrm{Var}[x_1+u_1]}\left(x_1+u_1-\mathrm{E}[x_1+u_1]\right) \\ =& \hat{\delta}_1 + \frac{\mathrm{Cov}[s, \beta_1 s+u_1]}{\mathrm{Var}[\beta_1 s+u_1]}\left(x_1+u_1-\mathrm{E}[\beta_1 s+u_1]\right) \\ =& \hat{\delta}_1 + \lambda_1\left(x_1+u_1\right), \end{aligned} \tag{C.3}$$

where

$$\lambda_1 \equiv \frac{\beta_1 \omega_s}{\beta_1^2 \omega_s + \omega_{u,1}}. \tag{C.4}$$

Given (C.3), the trader's problem (C.1) is rewritten as

$$\max_{x_1} \ \mathrm{E}\left[\left(\delta_1 - \left(\hat{\delta}_1 + \lambda_1(x_1+u_1)\right)\right)x_1 \middle| s\right] \ \Rightarrow \ \max_{x_1} \ sx_1 - \lambda_1 x_1^2. \tag{C.5}$$

The FOC for x_1 is $s - 2\lambda_1 x_1 = 0$, which determines the trader's optimal action:

$$x_1 = \frac{1}{2\lambda_1}s. \tag{C.6}$$

For the other agents' belief (C.2) to be consistent with (C.6), we need

$$\beta_1 = \frac{1}{2\lambda_1} \iff \beta_1 = \sqrt{\frac{\omega_{u,1}}{\omega_s}}, \tag{C.7}$$

which implies

$$\lambda_1 = \frac{1}{2}\sqrt{\frac{\omega_s}{\omega_{u,1}}}. \quad \square \tag{C.8}$$

D Proof of Lemma ??

Let us present $\delta_1 \sim N(\hat{\delta}_1 + s, \omega_\delta)$ equivalently as $\delta_1 = \hat{\delta}_1 + s + \epsilon_1$ with $\epsilon_1 \sim N(0, \omega_\delta)$. Then the trader's trading profit in $t = 1$ is

$$\begin{aligned}
\pi_1 &= (\delta_1 - p_1)x_1 \\
&= \left(\hat{\delta}_1 + s + \epsilon_1 - \left(\hat{\delta}_1 + \lambda_1\left(\beta_1 s + u_1\right)\right)\right)\beta_1 s \\
&= \left(\hat{\delta}_1 + s + \epsilon_1 - \left(\hat{\delta}_1 + \frac{1}{2}\sqrt{\frac{\omega_s}{\omega_{u,1}}}\left(\sqrt{\frac{\omega_{u,1}}{\omega_s}}s + u_1\right)\right)\right)\sqrt{\frac{\omega_{u,1}}{\omega_s}}s \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}s^2 + \left(\sqrt{\frac{\omega_{u,1}}{\omega_s}}\epsilon_1 - \frac{u_1}{2}\right)s. \qquad \text{(D.1)}
\end{aligned}$$

Taking the expectation of (D.1) from the recruiter's perspective, we obtain the trader's salary:

$$\begin{aligned}
w &= \mathrm{E}[\pi_1 | p_0, \delta_0] \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}\mathrm{E}[s^2 | p_0, \delta_0] \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}\left(\mathrm{E}[s | p_0, \delta_0]^2 + \mathrm{Var}[s | p_0, \delta_0]\right) \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}\left(\hat{s}^2 + \hat{\omega}_s\right), \qquad \text{(D.2)}
\end{aligned}$$

where $\hat{\omega}_s$ is the posterior variance of s, which is a constant. $\square$

E Proof of Propositions 3.1 and ??

In the following analyses, denote

$$a = \frac{\omega_{u,0}}{\omega_s},\ b = \frac{\omega_{u,0}}{\omega_\delta}, \tag{E.1}$$

so that we rewrite equation (3.15) as

$$\beta = F(\beta) \equiv \frac{1 + 2\theta\left(\frac{1}{a+b+\beta^2}\right)^2 b\beta}{2\left(\frac{\beta}{\beta^2+a} - \theta\left(\frac{\beta}{a+b+\beta^2}\right)^2\right)}. \tag{E.2}$$

By rearranging (E.2), the equilibrium β solves the following condition.

$$2\theta\beta = J(\beta^2), \tag{E.3}$$

where, by denoting $\beta^2 = y$,

$$J(y) = \frac{(y-a)\,(y+b+a)^2}{(y+a)\,(y+b)}. \tag{E.4}$$

The first-order derivative of the RHS of (E.3) with respect to β is

$$K^{(1)}(\beta) \equiv 2\beta J^{(1)}(\beta^2), \tag{E.5}$$

with

$$J^{(1)}(y) \equiv \frac{N(y)}{(y+a)^2\,(y+b)^2}, \tag{E.6}$$

where the numerator is

$$N(y) = y^4 + 2(a+b)y^3 + n_2 y^2 + n_1 y + n_0$$

with coefficients being

$$n_2 = 2a^2 + 6ab + b^2,$$

$$n_1 = 2a\left[3b^2 + 3ab + a^2\right],$$

$$n_0 = a\left(a^2 + ab + 2b^2\right)(a+b).$$

From (E.4) and (E.6), $y = a$ is the unique solution to $J(y) = 0$ in $y > 0$, and $J(y)$ is increasing in y for $y \geq a$. Hence, there is no equilibrium in $y < a$, and we focus on $y \geq a$.

The second-order derivative of the RHS of (E.3) with respect to β is expressed as

$$K^{(2)}(y) \equiv 2\left(J^{(1)}(y) + 2yJ^{(2)}(y)\right), \tag{E.7}$$

where $J^{(2)}$ denotes the second-order derivative of J with respect to y. $K^{(1)}$ and $K^{(2)}$ have the following properties:

Lemma E.1. *(i) $K^{(1)}(\beta)$ takes a U-shaped curve in relation to β.*
(ii) $K^{(2)}(a) < 0$ and $K^{(2)}(\infty) > 0$. There exists a unique solution to $K^{(2)}(y) = 0$.

Proof. Note that point (i) directly follows from point (ii). By denoting that $N^{(1)}(y) = \frac{dN(y)}{dy}$, it holds that

$$J^{(1)}(y) + 2yJ^{(2)}(y) = J^{(1)}(y)\frac{-7y^2 - 3(a+b)y + ab}{(y+a)(y+b)} + 2y\frac{\left(y^2 + (a+b)y + ab\right)N^{(1)}(y)}{(y+a)^3(y+b)^3}. \tag{E.8}$$

Evaluating at $y = a$,

$$\begin{aligned}
J^{(1)}(a) + 2aJ^{(2)}(a) &= -J^{(1)}(a)\frac{5a+b}{a+b} + \frac{N^{(1)}(y)}{2a\,(a+b)^2} \\
&\propto -2a(a+b)(5a+b)J^{(1)}(a) + N^{(1)}(a) \\
&= -4a^3 - b\,(4+b)\,a - b^3 \\
&< 0.
\end{aligned}$$

Therefore, $K^{(2)}(a) < 0$ holds.

The numerator of (E.8) is summarized as

$$T(y) = -\left[y^4 + 2(a+b)y^3 + n_2 y^2 + n_1 y + n_0\right]\left[7y^2 + 3(a+b)y - ab\right]$$
$$+ 2y\left(y^2 + (a+b)y + ab\right)\left[4y^3 + 6(a+b)y^2 + 2n_2 y + n_1\right].$$

The above argument suggests that $T(a) < 0$.

Using Mathematica, we summarize T into the following polynomial:

$$T(y) = \sum_{k=0}^{6} t_k y^k, \tag{E.9}$$

where the coefficients are

$$t_0 = a^2 b(a+b)\left(a^2 + ab + 2b^2\right) > 0,$$

$$t_1 = -3a\left(a^4 + (a+b)(a-b)^2 b + b^4\right) < 0,$$
$$t_2 = -3a\left(3a^3 + 4a^2 b + ab^2 + 5b^3\right) < 0,$$
$$t_3 = -8a^3 - 8a^2 b - 9ab^2 + b^3,$$
$$t_4 = 3b\,(a+b) > 0,$$
$$t_5 = 3(a+b) > 0,$$

and $t_6 = 1$. As $t_6 > 0$, $T(\infty) = \infty > 0$. Together with $T(a) < 0$, the intermediate value theorem implies that $T(y) = 0$ has at least one solution in $y > a$.

Suppose that y_{min} is one of the solutions to $T(y) = 0$. Consider the first-order derivative of T evaluated at $y = y_{min}$ and multiplied y_{min}:

$$\mathcal{T}(y_{min}) = y_{min}\frac{dT(y_{min})}{dy} = \sum_{k=1}^{6} k t_k y_{min}^k.$$

Applying $T(y_{min}) = 0$ and eliminating the terms related to t_3,

$$\mathcal{T}(y_{min}) = -3t_0 - 2t_1 y_{min} - t_2 y_{min}^2 + t_4 y_{min}^4 + 2t_5 y_{min}^5 + 3t_6 y_{min}^6,$$

According to the signs of t_k and $y_{min} > a$, it holds that

$$
\begin{aligned}
\mathcal{T}(y_{min}) &> -3t_0 - 2t_1 a - t_2 a^2 + t_4 a^4 + 2t_5 a^5 + 3t_6 a^6 \\
&= 3a^2 \left(6a^4 + 8a^3 b + b^4 + (a^2 - b^2)^2\right) \\
&> 0.
\end{aligned}
$$

Therefore, if y_{min} is a solution to $T(y) = 0$, then the slope of $T(y)$ at this point must be positive. Together with $T(a) < 0 < T(\infty)$, y_{min} is the unique solution to $T(y) = 0$, and $T(y) > 0 \Leftrightarrow y > y_{min}$.[22] The argument above implies that $K^{(1)}(\beta)$ takes a U-shaped curve with respect to β, and the minimum point is given by $\beta_{min} \equiv \sqrt{y_{min}}$. Note that $\tilde{\beta}$ in footnote **??** is given by $\tilde{\beta} = \beta_{min}$. □

The RHS of (E.3) is monotonically increasing in β, and Lemma E.1 attests that it is concave in $y \leq y_{min}$, while it becomes convex in $y > y_{min}$. In other words, the slope of the curve is the least steep at $y = y_{min}$, and it is given by

$$K_{min} = 2\beta_{min} J^{(1)}(y_{min}). \tag{E.10}$$

Note that K_{min} is represented only by parameters, a, b, and does not involve θ.

Consider the following condition to characterize equilibrium:

$$J(y_{min}) > K_{min}\beta_{min}. \tag{E.11}$$

Then the following result holds:

Lemma E.2. *(i) If inequality (E.11) holds, then there exist θ_H and θ_L with $\theta_H > \theta_L$. If, and only if, $\theta \in \Theta \equiv (\theta_L, \theta_H)$, then the equilibrium condition (E.3) has three solutions.*
(ii) If inequality (E.11) is violated, then $\theta_H = \theta_L$, and the equilibrium condition (E.3) has a unique solution. The solution is $\beta > \beta_{min}$ ($\beta < \beta_{min}$) if $\theta > \theta_H$

[22] Suppose that $T(y) = 0$ has more than two solutions. For any solutions, denoted as y_0 and y_1, the mean-value theorem implies that there exists at least one $y_2 \in [y_0, y_1]$ such that $\frac{dT(y_2)}{dy} = 0$. This contradicts to the strict positivity or negativity of $\mathcal{T}$.

($\theta < \theta_H$).

Proof. Consider the plot of the curve, $J(\beta^2)$, and the line, $2\theta\beta$, against β. With the curve being fixed, consider a line that goes through the origin and is tangent to the curve. The tangent point is characterized by β that satisfies

$$J(\beta^2) = K^{(1)}(\beta^2)\beta,$$

where the LHS represents the y-coordinate of the curve at β, and the RHS represents that of the line that goes through the origin. Note that their slopes must be identical and $K^{(1)}(\beta^2)$. Applying (E.5), the above condition is rewritten as

$$J(\beta^2) = 2\beta^2 J^{(1)}(\beta^2).$$

Hence, denoting $y = \beta^2$, the tangent point is characterized by the solution to

$$0 = G(y) = 2yJ^{(1)}(y) - J(y).$$

Note that we focus on $y > \sqrt{a}$, and thus y and β have one-to-one mapping.

Function $G(y)$ has the following first-order derivative:

$$\begin{aligned} G^{(1)}(y) &= 2\left(yJ^{(2)}(y) + J^{(1)}(y)\right) - J^{(1)}(y) \\ &= J^{(1)}(y) + 2yJ^{(2)}(y) \\ &= \frac{K^{(2)}(y)}{2}, \end{aligned}$$

where we apply the definition of $K^{(2)}$ in (E.7). Lemma E.1 has established that $K^{(2)}(a) < 0$ and $K^{(2)}(\infty) > 0$. Also, there is a unique $y = y_{min}$ such that $K^{(2)}(y_{min}) = 0$ and $K^{(2)}(y_{min}) > 0 \Leftrightarrow y > y_{min}$. Hence, $G(y)$ takes a U-shaped curve with respect to y and $y = y_{min}$ is the minimizer of the curve.

With $G(y)$ being U-shaped, $G(y) = 0$ has two solutions if and only if its minimum point dips below the x-axis, namely,

$$2y_{min}J^{(1)}(y_{min}) - J(y_{min}) < 0.$$

Applying the definition of K_{min} in (E.10), the above inequality is rewritten as condition (E.11).

When condition (E.11) holds and $G(y) = 0$ has two solutions, we denote them as y_0 and $y_1 (> y_0)$. The corresponding slopes of these tangent lines are $K^{(1)}(y_0)$ and $K^{(1)}(y_1)$, respectively. Then the original line, $2\theta\beta$, and the curve, $J(\beta^2)$, have three intersections if and only if

$$\frac{K^{(1)}(y_1)}{2} \equiv \theta_L < \theta < \theta_H \equiv \frac{K^{(1)}(y_0)}{2}. \tag{E.12}$$

This result concludes the proof of statement (i) of the lemma.

Conversely, if condition (E.11) does not hold, no tangent line that goes through the origin is obtained: function $J(\beta^2)$ becomes the least steep at $\beta_{min} = \sqrt{y_{min}}$, and the tangent line is specified by $K_{min}\beta - K_{min}\beta_{min} + J(\beta_{min}^2)$, while the intercept of this line is negative, as condition (E.11) is violated. Hence, the line, $2\theta\beta$, and the curve, $J(\beta^2)$ have a unique intersection. This intersection occurs in $\beta > \beta_{min}$ if, and only if, $\theta > \theta'_H \equiv \frac{J(\beta_{min}^2)}{2\beta_{min}}$.

To conclude statement (ii), we must show that θ_H and θ_L in statement (i) converges to θ'_H in statement (ii) at the threshold of these two cases so that we re-define it as $\theta_H = \theta_L = \theta'_H$. This result is shown in Lemma E.3 below. □

Finally, consider the comparative statics with respect to ω_s. The following result holds:

Lemma E.3. *(i) $J(y_{min}) - K_{min}\beta_{min}$ is increasing in ω_s. Hence, there exists a unique $\tilde{\omega}_s > 0$, such that, (E.11) holds with equality, and inequality (E.11) holds if and only if $\omega_s > \tilde{\omega}_s$.*
(ii) When condition (E.11) holds, both θ_H and θ_L increase as ω_s increases. Interval Θ expands, as the following inequalities hold:

$$0 < \frac{d\theta_L}{d\omega_s} < \frac{d\theta_H}{d\omega_s}.$$

(iii) At $\omega_s = \tilde{\omega}_s$, it holds that $\theta_H = \theta_L = \theta'_H$, while at the limit of $\omega_s \to 0$, $\theta'_H \to 0$.

(iv) There exists a unique threshold for φ^*, such that, $\varphi \geq \varphi^*(< \varphi^*)$ leads to case 1 (case 2) in Proposition **??**.

Proof of Statements (i) and (ii). Note that we are interested in the reaction of the following function when $a = \frac{\omega_{u,0}}{\omega_s}$ changes and, hereafter, we make the dependence of functions on a explicit:

$$2\theta_k = K^{(1)}(y_j, a) = 2\sqrt{y_j} J^{(1)}(y_j, a),$$

where $y_j = y_0, y_1$ denote the solutions to $G(y, a) = 0$, if any, and $k = L, H$ are appropriately determined following (E.12). According to the definition of G, the following partial derivative holds.

$$\frac{\partial G(y, a)}{\partial a} = 2y \frac{\partial J^{(1)}(y, a)}{\partial a} - \frac{\partial J(y, a)}{\partial a}.$$

The first term is summarized by

$$\begin{aligned} \frac{\partial J^{(1)}(y, a)}{\partial a} &\propto (y + a) \frac{\partial N(y, a)}{\partial a} - 2N(y, a) \\ &= (y + a) \left[2y^3 + n_{2,a} y^2 + n_{1,a} y + n_{0,a}\right] - \left[y^4 + 2(a + b)y^3 + n_2 y^2 + n_1 y + n_0\right] \\ &= \sum_{k=0}^{4} A_k y^k \end{aligned}$$

where $A_4 = 1$, $A_3 = 4(a + b)$,

$$A_2 = 2a^2 + 5b^2 + 2(3ab + a^2) + 2a\,[3b + 2a]\,,$$

$$A_1 = 2a^2\,[3b + 2a] \left(a^2 + ab + 2b^2\right)(2a + b) + a\,(2a + b)\,(a + b),$$

$$\begin{aligned} A_0 &= a\left[(2a + b)\left(a^2 + ab + 2b^2 + a(a + b)\right) - \left(a^2 + ab + 2b^2\right)(a + b)\right] \\ &> (a(a + b))^2\,. \end{aligned}$$

Hence, $\frac{\partial J^{(1)}(y,a)}{\partial a} > 0$. The second term is

$$(y+b)\frac{\partial J(y,a)}{\partial a} = -2\frac{(y+b+a)\left(y(a+b)+a^2\right)}{(y+a)^2}$$
$$< 0. \tag{E.13}$$

Therefore, $\frac{\partial G}{\partial a} > 0$. As the previous proof has established that $G(y,a)$ takes a U-shaped curve, $\frac{\partial G}{\partial a} > 0$ implies that its minimum point is monotonically decreasing in ω_s, concluding the first statement of point (i) in the lemma.

As $a \to 0$, it holds that $y_{min} \to 0$ and

$$\lim_{a\to 0} G(y_{min}, a) = \lim_{a\to 0}[2y_{min} - (y_{min} + b)] = -b < 0.$$

Also

$$\lim_{a\to\infty} G(y_{min}, a) = +\infty.$$

Therefore, together with

$$\frac{dG(y_{min}, a)}{da} = \frac{dy_{min}}{da}\frac{\partial G}{\partial y} + \frac{\partial G}{\partial a} > 0,$$

the above limits imply the existence of $\bar{a}$, such that, condition (E.11) holds if and only if $a < \bar{a}$. Given $\omega_{u,0}$, this result can be translated into ω_s so that $\tilde{\omega}_s = \frac{\omega_{u,0}}{\bar{a}}$. Noting that $a < \bar{a} \Leftrightarrow \omega_s > \tilde{\omega}_s$, we conclude the second half of point (i) in the lemma.

Suppose that condition (E.11) holds, meaning that y_0 and y_1 exist. For $j \in 0, 1$, $\frac{\partial G}{\partial a} > 0$ implies that $\frac{dy_0}{da} > 0$ and $\frac{dy_1}{da} < 0$. Also, the implicit function theorem leads to

$$\frac{dy_j}{da} = -\frac{1}{G^{(1)}(y_j, a)}\frac{\partial G(y_j, a)}{\partial a}.$$

By taking the total derivative of θ_k,

$$\begin{aligned}\frac{d\theta_k}{da} &= \frac{1}{2}\frac{1}{\sqrt{y_j}}\frac{dy_j}{da}G^{(1)}(y_j,a) + \sqrt{y_j}\frac{\partial J^{(1)}(y_j,a)}{\partial a} \\ &= -\frac{1}{2}\frac{1}{\sqrt{y_j}}\left(2y_j\frac{\partial J^{(1)}(y_j,a)}{\partial a} - \frac{\partial J(y_j,a)}{\partial a}\right) + \sqrt{y_j}\frac{\partial J^{(1)}(y_j,a)}{\partial a} \\ &= \frac{1}{2}\frac{1}{\sqrt{y_j}}\frac{\partial J(y_j,a)}{\partial a} \\ &< 0,\end{aligned} \tag{E.14}$$

where the last inequality follows from (E.13). Therefore, we conclude that the boundary slopes, θ_H and θ_L are both increasing in ω_s.

Consider the magnitude of the reactions. By plugging (E.13) into (E.14),

$$\frac{d\theta_k}{da}| \propto \frac{\sqrt{y_j}}{(y_j+a)^2}\left(1+\frac{a}{y_j+b}\right)\left(1+\frac{1}{y_j}\frac{a^2}{a+b}\right).$$

It is straightforward that the last terms are all decreasing in y_j. Since we compare $y_0 < y_1$ with a being fixed, we obtain $\frac{d\theta_H}{d\omega_s} > \frac{d\theta_L}{d\omega_s}$, thereby concluding the proof.

Proof of Statements (iii) and (iv). Note that $\lim_{\omega_s\to\tilde{\omega}_s} y_j = y_{min}$, and $J(y_{min}) = K_{min}\beta_{min} = 2\beta_{min}J^{(1)}(y_{min})$ holds. Therefore, at this limit,

$$\theta'_H = \frac{J(\beta_{min}^2)}{2\beta_{min}} = \beta_{min}J^{(1)}(\beta_{min}^2) = \frac{K_{min}}{2} = \theta_k,$$

where the last equality holds because of (E.5). This concludes the proof of statement (iii).

Finally, consider the limit at $\omega_s \to 0$, which implies $a \to \infty$ (given $\omega_{u,0}$). Since $y_{min} \searrow a$, we can denote it as $y_{min} = a + \epsilon$, where $\epsilon > 0$ is a small positive and $\lim_{\omega_s\to\tilde{\omega}_s}\epsilon = 0$. With this notation, function J is rewritten at the limit as

$$\lim_{\omega_s\to\tilde{\omega}_s} J(y_{min},a) = \lim_{a\to\infty,\epsilon\to 0}\frac{\epsilon(2a+b+\epsilon)}{(2a+\epsilon)(a+b+\epsilon)} = 0.$$

Since $\theta'_H = \frac{J(y_{min})}{2\sqrt{y_{min}}}$, we conclude that $\lim_{\omega_s \to \tilde{\omega}_s} \theta'_H = 0$.

As for statement (iv), case 1 occurs if, and only if, the curve $\theta = \frac{\varphi}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}$ exceeds θ_H at $\omega_s = \tilde{\omega}_s$. The above analyses have already established that $\theta_H = \frac{K^{(1)(y_{min})}}{2}$, where y_{min} is implicitly characterized by parameters a and b, we can define the threshold φ^* as

$$\varphi^* \equiv K^{(1)(y_{min})}\sqrt{\frac{\omega_{u,0}}{\bar{a}\omega_{u,1}}}. \tag{E.15}$$

□

Note that Proposition 3.1 follows from points (i) and (ii) of Lemma E.3, whereas Proposition **??** is shown by putting monotonically decreasing θ in equation (**??**) together with points (i) and (ii).

F Proof of Lemma ?? and Proposition 4.1

The optimal trading strategy satisfies the following condition.

$$\therefore (1 - 2\lambda_0\beta_0) + \frac{\psi}{2\lambda_1}\xi^2(\gamma^2\beta_0 + \gamma(1-\gamma)) = 0. \tag{F.1}$$

The marginal benefit of increasing s^* is given by

$$\mathrm{E}_\beta[v(\omega_s, \beta_0)] = \left[(1 - \lambda_0\beta_0) + \frac{\psi}{4\lambda_1}\xi^2(\gamma^2\beta_0 + 2\gamma(1-\gamma))\right]\beta_0 + \frac{\psi}{4\lambda_1}\xi^2(1-\gamma)^2 + (1-\psi)\frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}. \tag{F.2}$$

Then, it holds that

$$\begin{aligned}\frac{\partial \mathrm{E}_\beta[v(\omega_s, \beta_0)]}{\partial \beta_0} &= \left[(1 - 2\lambda_0\beta_0) + \frac{\psi}{2\lambda_1}\xi^2(\gamma^2\beta_0 + \gamma(1-\gamma))\right] \\ &+ \left[-\beta_0\frac{\partial\lambda_0}{\partial\beta_0} + \frac{\psi}{4\lambda_1}\frac{\partial}{\partial\beta_0}\left[\xi^2(\gamma^2\beta_0 + 2\gamma(1-\gamma))\right]\right]\beta_0 + \frac{\psi}{4\lambda_1}\frac{\partial}{\partial\beta_0}\left(\xi^2(1-\gamma)^2\right).\end{aligned}$$

Note that the first term is zero due to the envelope theorem using (F.1). By using the notations introduced in equations (E.1), the remaining terms are summarized as follows.

$$-\beta_0\frac{\partial}{\partial\beta_0}\frac{\beta_0}{\beta_0^2+a}+\frac{\psi}{4\lambda_1}\left[\beta_0^2\frac{\partial}{\partial\beta_0}\left(\frac{\beta_0}{a+b+\beta_0^2}\right)^2+2\beta_0\frac{\partial}{\partial\beta}\left(\frac{1}{a+b+\beta_0^2}\right)^2 b\beta_0+\frac{\partial}{\partial\beta}\left(\frac{b}{a+b+\beta_0^2}\right)^2\right]. \tag{F.3}$$

From the proof in Appendix E, it holds that $y > a$ in the equilibrium. Therefore, the first term is negative. As for the second term, it holds that

$$\begin{aligned}\text{Second terms of (F.3)} &= 2\beta_0^3\left(\frac{a+b-\beta_0^2}{(a+b+\beta_0^2)^3}\right)+2\frac{a+b-3\beta_0^2}{(a+b+\beta_0^2)^3}b\beta_0-4b^2\frac{\beta_0}{(a+b+\beta_0^2)^3}\\ &\sim -y^2+y(a+b-3b)+b(a-b).\end{aligned}$$

Since $y > a$, the quadratic equation above is negative for all $y > a$ if and only if it is negative at $y = a$. This is true because

$$-a^2+a(a-2b)+b(a-b)=-ab-b^2<0.$$

Therefore, the marginal benefit in (F.2) is decreasing in β_0, concluding the proof of Lemma **??**. Proposition 4.1 directly follows from Lemma **??** and equation (**??**).